Robot Vision: Mock Exam 4 (Solutions)

Chapters 2 to 12 · 90 minutes · no documents · 40 points

Part A: Multiple choice

22 points, about 35 min

A1 Which statement about activation functions is correct? (ch. 2)

(a) ReLU outputs 0 for positive inputs.
(b) The sigmoid is the best activation for every layer.
(c) ✓ **ReLU does not saturate for positive inputs, while sigmoid and tanh saturate for large $|u|$.**
(d) GELU is mainly used in early CNNs such as LeNet.

GELU is common in Transformers; no activation is best everywhere.

A2 What is the main difference between full-batch gradient descent and mini-batch SGD? (ch. 2)

(a) Full-batch gradient descent uses one sample per update.
(b) ✓ **Full-batch uses the whole training set for each update; mini-batch SGD estimates the gradient from a small subset.**
(c) There is no difference in what each update uses.
(d) Mini-batch SGD uses second-order derivatives.

Mini-batch: cheaper steps, and the noise can help.

A3 Which statement about the vanishing gradient is correct? (ch. 3)

(a) ✓ **Small derivatives of saturating activations multiply across layers; ReLU helps for positive inputs.**
(b) It only happens in networks with one hidden layer.
(c) The sigmoid derivative can reach 1, so it never vanishes.
(d) ReLU causes the vanishing gradient.

Sigmoid derivative ≤ 0.25.

A4 What is the main idea of the Adam optimizer? (ch. 3)

(a) It uses the exact Hessian.
(b) It only accumulates squared gradients, like AdaGrad.
(c) It uses a fixed step for every weight.
(d) ✓ **It uses moving averages of both the gradients and the squared gradients to set each step.**

Momentum + RMSProp, with bias correction.

A5 How is one output value of a convolutional layer computed? (ch. 4)

(a) Take the maximum of the local region.
(b) Average all the pixels of the image.
(c) Multiply all the values of the region together.
(d) ✓ **Multiply the input values in a local region by the filter weights and sum the products.**

The maximum is max pooling.

A6 A convolutional layer receives K input channels and has M filters. Which statement is correct? (ch. 4)

(a) The output always has $K \times M$ channels.
(b) Each filter sees one input channel; the output has K channels.
(c) ✓ **Each filter has K channels, and the output has M channels.**
(d) Each filter has M channels, and the output has K channels.

H^2KM weights + M biases.

A7 Which normalization is stable with very small batches? (ch. 5)

(a) ✓ **Group normalization, because it computes statistics within each example.**
(b) Batch normalization with a batch of size 1.

(c) Batch normalization, because it uses the mini-batch statistics.
(d) None, all need large batches.

BN is unstable with small batches.

A8 What is a key difference between bilinear interpolation and transposed convolution? (ch. 5)

(a) Transposed convolution inverts the convolution exactly.
(b) Both use learned weights.
(c) Bilinear interpolation reduces the resolution.
(d) **✓ Bilinear interpolation uses fixed weights given by positions; transposed convolution uses learned weights.**

Transposed conv applies $A^\top$, not A^{-1}.

A9 How does Network Dissection decide that a unit detects a concept? (ch. 6)

(a) By training a linear classifier on the unit.
(b) By removing the unit and measuring the accuracy.
(c) By the gradient of the class score with respect to the unit.
(d) **✓ By the IoU between the unit's thresholded, upsampled activation map and the segmentation of the concept.**

Overlap with concept masks.

A10 How does LIME explain a prediction? (ch. 6)

(a) It computes the exact gradient at every pixel.
(b) **✓ It samples on/off versions of superpixels and fits a sparse local linear model around the input.**
(c) It averages over all subsets of features.
(d) It retrains the network without the image.

The last option describes Shapley values.

A11 How is a pretrained CNN used as a fixed feature extractor? (ch. 7)

(a) Its first layer is replaced by a new one.
(b) Its features are discarded.
(c) **✓ Its weights are frozen and only a new classifier is trained on its features.**
(d) All its weights are retrained from scratch.

Linear probing.

A12 Under the standard multiple instance learning assumption, when is a bag positive? (ch. 7)

(a) When it contains exactly one instance.
(b) **✓ When at least one of its instances is positive.**
(c) When all its instances are positive.
(d) When most of its instances are positive.

Aggregation must be permutation-invariant.

A13 How are positive pairs built in SimCLR? (ch. 8)

(a) **✓ Two different augmented views of the same image.**
(b) Two random images from the dataset.
(c) An image and its caption.
(d) Two images of the same labeled class.

No labels needed.

A14 Why does structured pruning speed up inference on GPUs better than unstructured pruning? (ch. 8)

(a) **✓ It removes whole filters or channels, so the remaining computations are smaller dense operations.**
(b) It quantizes the weights to 8 bits.

(c) It removes more individual weights.
(d) It keeps the irregular sparsity that GPUs exploit.

Unstructured sparsity is irregular.

A15 Which statement about a TCN with causal dilated convolutions is correct? (ch. 9)

(a) Each output depends on the whole future sequence.
(b) Its receptive field is infinite.
(c) **✓ Each output depends on a finite window of past inputs, and positions can be computed in parallel when the sequence is known.**
(d) It must process the time steps one by one during training.

Field $= 1 + (k-1)\sum d$.

A16 How can a GNN node use information from nodes several edges away? (ch. 9)

(a) One layer already sees the whole graph.
(b) By sorting the nodes.
(c) **✓ Each layer aggregates the neighbors, so stacking k layers reaches nodes k edges away.**
(d) By using a different weight for each node.

One layer = one hop.

A17 What is the essential idea of self-attention? (ch. 10)

(a) Tokens of two different sequences exchange information.
(b) Each token is processed independently.
(c) **✓ Tokens of the same sequence exchange information according to their relationships.**
(d) Tokens are processed one after the other.

No recurrence, so it is parallel.

A18 Why did ViT benefit more from large-scale pre-training than CNNs? (ch. 10)

(a) It has fewer parameters than CNNs.
(b) It cannot be trained on small images.
(c) It has stronger inductive biases than CNNs.
(d) **✓ It has weaker built-in spatial inductive biases, so it must learn spatial structure from data.**

With little data, CNNs win.

A19 Why does VQA v2.0 pair a question with visually similar images that have different answers? (ch. 11)

(a) To double the number of images.
(b) **✓ To reduce the shortcut of answering from the question alone.**
(c) To make the questions longer.
(d) To train without labels.

Language priors.

A20 How is CLIP trained? (ch. 11)

(a) By reconstructing the image from the text.
(b) By classifying ImageNet images.
(c) **✓ Contrastively: matching image-text pairs (the diagonal of the similarity matrix) are made similar, others dissimilar.**
(d) By generating captions word by word.

About 400M web image-text pairs.

A21 What is mode collapse in a GAN? (ch. 12)

(a) **✓ The generator produces only a few kinds of similar images and loses the diversity of the data.**
(b) The discriminator always outputs 0.5.
(c) The generator copies the training images exactly.

(d) The latent space becomes too large.

One realistic image can be enough to fool D.

A22 How is a DDPM diffusion model trained? (ch. 12)

(a) Train a discriminator on noisy images.
(b) **✓ Sample one time step t and a noise, build x_t, and train the network to predict the noise (squared error).**
(c) Run the full reverse chain for each image.
(d) Maximize the exact likelihood with a Jacobian.

The full chain runs only for sampling.

Part B: Short answers

12 points, 3 per question, about 35 min

B1 In activation maximization, what is optimized, and why is a regularizer needed? (ch. 6)

The input image is optimized to maximize a unit or a class score. Without a regularizer the result is high-frequency noise; $R(x)$ keeps it looking like a natural image.

B2 You have few labeled images and an ImageNet-pretrained CNN. Which layers do you freeze, and why? (ch. 7)

Freeze the early layers: their features (edges, textures) are general, and fewer trainable parameters means less overfitting with few labels. Train a new head, possibly fine-tune the upper layers.

B3 Explain aleatoric and epistemic uncertainty with one example each. Which one can more data reduce? (ch. 8)

Aleatoric: noise in the data (an image too blurry). Epistemic: lack of knowledge (too few examples of a case). More data reduces only epistemic uncertainty.

B4 Explain the difference between self-attention and cross-attention, with an example from VQA. (ch. 11)

Self-attention: queries, keys and values from the same sequence (question words attend to each other). Cross-attention: queries from one sequence, keys and values from another (question tokens query the image tokens).

Part C: Calculations

6 points, 2 per question, about 15 min

C1 A 28×28 image goes through a 5×5 convolution (stride 1, no padding), then 2×2 max pooling with stride 2. Output spatial size? (ch. 4)

Conv: $28 - 5 + 1 = 24$. Pooling: $\mathbf{12} \times \mathbf{12}$.

C2 ViT cuts a 224×224 image into 16×16 patches and adds a class token. How many tokens enter the encoder? (ch. 10)

$(224/16)^2 = 14^2 = 196$ patches $+1$ class token $= \mathbf{197}$.

C3 MC dropout runs 10 passes on an image: cat 7 times, dog 2 times, fox once. What is the variation ratio? (ch. 8)

$1 - 7/10 = \mathbf{0.3}$.